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COURSE PACK

## Syllabus

# Introduction to Digital Forensics

A 14-week undergraduate course: Windows artifacts, memory, network and mobile evidence, ending in a graded capstone.

| | |
|---|---|
| **Course code** | [DEPT 000] |
| **Term** | [Term and year] |
| **Instructor** | [Name] |
| **Contact** | [Email] |
| **Office hours** | [Days, times, location or link] |
| **Meeting pattern** | [Days, times, room or lab] |
| **Length** | 14 weeks, one lab per week |
| **Discipline** | Digital Forensics |
| **Level** | Undergraduate, second or third year |
| **Prerequisites** | Operating systems fundamentals and basic networking. No prior forensics. |
| **Lab time** | About 13 hours of hands-on work across the term |
| **Prepared** | 2026-11-13 |

**Bracketed fields are for the adopting instructor to complete. Everything else can be used as written. Every lab in this course generates a different variant of its case for each student, with an answer key verified against that student's own evidence; grading and the student portal run on the instructor's machine.**

# 1. Course description

Students in this course learn digital forensics by doing it: they receive evidence, work it with the free tools examiners use on the job, and write up what they found. The first six weeks each isolate one Windows artifact family (removable devices, program execution, user files, persistence, event logs) and end by combining them into a timeline. A timed midterm applies all six to an insider theft. The second half widens the evidence to browser, cloud, memory, network and mobile sources, and the course closes with a timed capstone on a credential theft traced from one workstation and a written investigation report. Every lab is a different variant of its case for every student, so there's no answer key to copy; each student has to read their own evidence.

# 2. Learning outcomes

On completing the course a student can:

1. Acquire and verify evidence, and keep a chain of custody.
2. Recover program execution, file, USB and persistence artifacts from a Windows system.
3. Build and defend a timeline from several artifacts.
4. Triage memory, network and mobile evidence with free tools.
5. Write an investigation report that states findings, cites artifacts and explains method.

# 3. Prerequisites and materials

## Prerequisites

Operating systems fundamentals and basic networking. No prior forensics.

## Software

All tools are free. One-run install scripts for each are on the Tools page of the product website.

| Tool | Used for |
|---|---|
| Autopsy | disk image triage and file recovery |
| Registry Explorer | SYSTEM, SOFTWARE and NTUSER hives |
| EvtxECmd | Windows event logs to CSV |
| PECmd | Prefetch |
| LECmd | LNK shortcut files |
| JLECmd | jump lists |
| Volatility 3 | memory images |
| Wireshark | packet captures |
| DB Browser for SQLite | browser and mobile databases |
| Timesketch | super-timelines (optional; a spreadsheet works) |

## Hardware

A laptop or lab workstation with 8 GB of RAM (16 GB for memory forensics weeks), 20 GB of free disk and permission to install the listed tools. Windows, macOS and Linux all work; some tools are Windows-only and run in a virtual machine elsewhere.

## Evidence

Evidence is generated per student and handed out as a ZIP or through a private portal link each week. No textbook purchase is required; each week names its reading from the free references below.

## Readings and references

No purchase is required. The texts below are free from their publishers, and each week names the part to read. Editions and addresses change; the title and publisher are the stable reference.

| Title | Publisher | Use in this course |
|---|---|---|
| Guide to Integrating Forensic Techniques into Incident Response (SP 800-86) https://doi.org/10.6028/NIST.SP.800-86 | NIST | The course spine for process: collection, examination, analysis, reporting. |
| Forensic Examination of Digital Evidence: A Guide for Law Enforcement https://nij.ojp.gov/library/publications/forensic-examination-digital-evidence-guide-law-enforcement | US National Institute of Justice | Chain of custody and documentation, weeks 1 and 14. |
| Windows Forensic Analysis poster https://www.sans.org/posters/ | SANS Institute (free posters) | One page per artifact family; the map for weeks 1 to 6 and 8. |
| Windows security auditing: events to monitor https://learn.microsoft.com/windows-server/identity/ad-ds/plan/appendix-l--events-to-monitor | Microsoft Learn | Event id reference for week 5 and the midterm. |
| Eric Zimmerman's tools documentation https://ericzimmerman.github.io/ | Eric Zimmerman | PECmd, LECmd, JLECmd, EvtxECmd, Registry Explorer and Timeline Explorer usage. |
| Autopsy user documentation https://sleuthkit.org/autopsy/docs/user-docs/ | Sleuth Kit Labs | Disk image triage, weeks 1, 7 and 14. |
| Volatility 3 documentation https://volatility3.readthedocs.io/ | Volatility Foundation | Memory forensics, weeks 10 and 11. |
| Wireshark User's Guide https://www.wireshark.org/docs/wsug_html_chunked/ | Wireshark Foundation | Display filters and following streams, week 12. |
| Guidelines on Mobile Device Forensics (SP 800-101 Rev. 1) https://doi.org/10.6028/NIST.SP.800-101r1 | NIST | Mobile data stores and extraction levels, week 13. |
| Open Textbook Library: digital forensics titles https://open.umn.edu/opentextbooks/ | University of Minnesota | Search for current open, CC-licensed forensics texts to assign as background reading. |

# 4. How a week runs

1. Each week has one lab built on a single case. Every student receives a different variant of that case: different names, hosts and timestamps, and most answers, generated from the same story.

2. Before the lab, read the briefing and the week's preparation notes, and confirm the listed tools open on your machine.

3. During the lab, work the evidence with the listed tools and record each answer with the artifact and method that proves it.

4. Submit through the student portal link for that week, or hand in the worksheet. Portal submissions are graded against your own answer key the moment you submit.

5. After release, review the walkthrough for your variant: where each answer was, and how it was found.

# 5. Assessment and grading

| Component | Weight | Details |
|---|---|---|
| Weekly labs (12) | 50% | Best 10 of 12 count. Auto-graded on submission; each lab is up to 20 questions on that week's case. |
| Midterm practical | 20% | Week 7. A 75-minute timed exam in the portal on a full insider-theft case covering weeks 1 to 6. |
| Capstone practical | 20% | Week 14. A 120-minute timed exam on a credential theft traced from one workstation. |
| Investigation report | 10% | Due one week after the capstone. A written report on the capstone case, graded with the rubric below. |

## Grading scale

| | |
|---|---|
| **A** | 90 to 100 |
| **B** | 80 to 89 |
| **C** | 70 to 79 |
| **D** | 60 to 69 |
| **F** | below 60 |

Timed assessments are graded on submission in the portal. Reports are graded with the rubric in section 8. Scores export to the institution's LMS from each week's job.

# 6. Course policies

## Collaboration

Discussing tools and technique with classmates is encouraged. Answers aren't transferable: each variant has its own evidence and its own key, and a submission that matches another student's variant rather than your own is treated as academic misconduct.

## Late work

Labs are due at the end of the week they're set. Late labs lose 10% per day for three days and aren't accepted after that. The lowest lab scores are dropped as stated in the assessment table, which covers illness and emergencies without paperwork for short absences.

## Timed assessments

Timed exams run in the student portal with a fixed clock, shuffled question order and integrity logging (tab switches, full-screen exits and pastes are recorded). Arrive with your tools installed and tested.

## Resubmission

One lab per term may be resubmitted within a week of feedback for up to 80% of full credit.

## Evidence handling

Evidence files are synthetic. Treat them as you would real evidence anyway: verify hashes, work on copies, and document every step. That habit is assessed.

## Accommodations

[Insert the institution's disability accommodation and extended-time statement. The portal supports per-student time extensions.]

## Academic integrity

[Insert the institution's academic integrity statement.]

# 7. Schedule

One case per week. The mode and time budget are set when the course is generated; the deliverable is what the student hands in.

| Wk | Topic | Case | Mode | Assessment |
|---|---|---|---|---|
| 1 | Evidence handling and the registry | First Look: USB Device History | Practice, about 33 min | Lab 1 |
| 2 | Program execution | First Look: What Ran on This Machine | Practice, about 33 min | Lab 2 |
| 3 | Files the user touched | First Look: Recent Files and Jump Lists | Practice, about 33 min | Lab 3 |
| 4 | Persistence | First Look: What Runs at Startup | Practice, about 33 min | Lab 4 |
| 5 | Windows event logs | First Look: Accounts and Services in the Logs | Practice, about 35 min | Lab 5 |
| 6 | Timelines | First Look: Building a Timeline | Practice, about 37 min | Lab 6 |
| 7 | Midterm practical | Insider Data Theft via USB | Timed exam, 75 min | Midterm 20% |
| 8 | Browser and download forensics | First Look: Suspicious Download | Practice, about 33 min | Lab 7 |
| 9 | Data exfiltration | The Customer Export | Practice, about 80 min | Lab 8 |
| 10 | Memory forensics | First Look: What's in Memory | Practice, about 37 min | Lab 9 |
| 11 | Code injection in memory | Injected Code in Memory | Practice, about 119 min | Lab 10 |
| 12 | Network forensics | First Look: Reading a Packet Capture | Practice, about 33 min | Lab 11 |
| 13 | Mobile forensics | The Texts They Deleted | Practice, about 70 min | Lab 12 |
| 14 | Capstone practical and report | Credential Theft and Lateral Movement | Timed exam, 120 min, with report | Capstone 20% + report 10% |

# Week by week

Each week below lists its objectives, the case and what it contains, what students should do beforehand, what they hand in, and a note for the instructor.

## Week 1: Evidence handling and the registry

**Objectives**

• Explain acquisition, hashing and chain of custody.

• Read USBSTOR and device properties in a SYSTEM hive.

**Case**

First Look: USB Device History. Registry hives and a shortcut file. Work out which USB drive was plugged into the machine, when, and what was opened from it.

Practice, about 33 min · Up to 15 questions · Tools: Registry Explorer, RegRipper, LECmd

**Reading**

SP 800-86 sections 3.1 to 3.2 (collection) and the NIJ guide chapter on documenting the scene. SANS poster: USB devices.

**Before the lab**

Install Registry Explorer and Autopsy. Print the chain-of-custody form from the course materials.

**Deliverable**

Lab 1 answers in the portal and a completed chain-of-custody form for the evidence ZIP.

**Instructor note**

Spend the first half hour on hashing and the custody form before anyone opens a hive; it sets the standard for the term.

## Week 2: Program execution

**Objectives**

• Interpret Prefetch and UserAssist.

• Tell a decoy execution from the real one.

**Case**

First Look: What Ran on This Machine. One user's execution artifacts. Test a user's claim that they never ran a program they downloaded.

Practice, about 33 min · Up to 15 questions · Tools: PECmd, Registry Explorer, LECmd

**Reading**

SANS poster: program execution (Prefetch and UserAssist). PECmd page in the Zimmerman docs.

**Before the lab**

Install PECmd. Read the program execution section of the SANS poster.

**Deliverable**

Lab 2 answers. For one question, a screenshot of the record that proves it.

**Instructor note**

Notepad sits in Prefetch beside the utility as ordinary activity; the discussion of how students ruled it out is the lesson.

## Week 3: Files the user touched

**Objectives**
• Parse LNK files and jump lists.

• Recover a target path and its timestamps.

**Case**
First Look: Recent Files and Jump Lists. Shortcuts, jump lists and registry MRUs. Reconstruct what a departing employee opened, where each file lived, and when.

Practice, about 33 min · Up to 15 questions · Tools: JLECmd, LECmd, Registry Explorer

**Reading**
SANS poster: file and folder opening (LNK, jump lists). LECmd and JLECmd pages.

**Before the lab**
Install LECmd and JLECmd.

**Deliverable**
Lab 3 answers.

**Instructor note**
The drive type and volume serial on one LNK show the file was opened from removable media; have students name the field that proves it.

## Week 4: Persistence

**Objectives**
• Enumerate Run keys and services.

• Rank persistence mechanisms by stealth.

**Case**
First Look: What Runs at Startup. A workstation keeps reinfecting itself after cleanup. Find what brings the malware back.

Practice, about 33 min · Up to 15 questions · Tools: Registry Explorer, RegRipper, PECmd

**Reading**
SANS poster: persistence. Microsoft Learn: Run and RunOnce keys, scheduled tasks.

**Before the lab**
Read the persistence section of the SANS poster: Run keys and services.

**Deliverable**
Lab 4 answers and a one-paragraph ranking of the persistence mechanisms found, most to least stealthy.

**Instructor note**
Ask for the ranking before revealing the key; it's the first judgment call of the course.

## Week 5: Windows event logs

**Objectives**

• Read 4720 and 7045 events.

• Distinguish the acting account from the target account.

**Case**

First Look: Accounts and Services in the Logs. Exported event logs from one server. Find the account and service changes made overnight and who made them.

Practice, about 35 min · Up to 15 questions · Tools: Event Viewer, EvtxECmd

**Reading**

Microsoft events to monitor: 4720, 4732, 7045, 1102. EvtxECmd page.

**Before the lab**

Install EvtxECmd and Timeline Explorer. Skim the Microsoft events to monitor table for 4720 and 7045.

**Deliverable**

Lab 5 answers.

**Instructor note**

The acting-account versus target-account confusion in the 4720 catches most students; make it an explicit question in class.

## Week 6: Timelines

**Objectives**

• Normalize timestamps across artifacts.

• Build and present a super-timeline.

**Case**

First Look: Building a Timeline. Two artifacts, one clock. Line up a browser download and a Security log to reconstruct the order things happened.

Practice, about 37 min · Up to 15 questions · Tools: Timesketch, EvtxECmd, DB Browser for SQLite

**Reading**

SP 800-86 section 3.3 (examination and analysis). Timeline Explorer page.

**Before the lab**

Choose a timeline tool: Timesketch, Timeline Explorer or a spreadsheet. Have weeks 1 to 5 output at hand.

**Deliverable**

Lab 6 answers and a one-page super-timeline of the week's case in one timezone.

**Instructor note**

A good checkpoint before the midterm: the timeline shows who can combine artifacts and who is still working one at a time.

## Week 7: Midterm practical

**Objectives**
• Apply weeks 1 to 6 to a full insider-theft case under time.

**Case**
Insider Data Theft via USB. An engineer who has resigned is suspected of walking out with confidential files on a USB stick. Prove what was connected, what was copied, and when.

Timed exam, 75 min · Up to 20 questions · Tools: Autopsy, Registry Explorer, FTK Imager, PECmd, JLECmd, LECmd

**Reading**
Review weeks 1 to 6. No new reading.

**Before the lab**
Review weeks 1 to 6. Test every tool the night before; the clock doesn't stop for installs.

**Deliverable**
Midterm submitted in the portal within 75 minutes.

**Instructor note**
Run in exam mode with integrity logging. Release scores after the session so late sitters aren't tipped off.

## Week 8: Browser and download forensics

**Objectives**
• Trace a download from history to disk.

• Read browser SQLite stores.

**Case**
First Look: Suspicious Download. One Chrome profile. Work out what the user searched for, where they went, and what they downloaded.

Practice, about 33 min · Up to 15 questions · Tools: DB Browser for SQLite, Browser History Viewer

**Reading**
SANS poster: browser usage. Autopsy user docs: web artifacts.

**Before the lab**
Install DB Browser for SQLite. Locate the browser history database on your own machine first.

**Deliverable**
Lab 7 answers.

**Instructor note**
Have students find their own download history before the lab; the structure then feels familiar.

## Week 9: Data exfiltration

**Objectives**
• Correlate browser, LNK and email evidence.

• Prove which file left and when.

**Case**
The Customer Export. A departing sales manager's workstation pulled a full customer export from the CRM. Find out where it went.

Practice, about 80 min · Up to 20 questions · Tools: Browser History Viewer, DB Browser for SQLite, LECmd, Registry Explorer, Thunderbird

**Reading**
SP 800-86 section 3.4 (reporting). Review the browser and LNK sections of the poster.

**Before the lab**
Review the browser and LNK sections of the SANS poster.

**Deliverable**
Lab 8 answers and a short paragraph proving which file left and when, citing three artifacts.

**Instructor note**
This is the first lab where no single artifact answers the question; grade the paragraph on correlation, not length.

## Week 10: Memory forensics

**Objectives**
• Run Volatility 3 process, network and DLL plugins.

• Identify the suspicious process.

**Case**
First Look: What's in Memory. One memory capture. Find the process that shouldn't be running, where it lives, and the server it phones home to.

Practice, about 37 min · Up to 15 questions · Tools: Volatility, text editor

**Reading**
Volatility 3 docs: getting started, pslist, pstree, netscan, dlllist.

**Before the lab**
Install Volatility 3 and run windows.info against the memory image in the lab pack while online, so it caches the Windows symbols. 16 GB of RAM recommended.

**Deliverable**
Lab 9 answers.

**Instructor note**
Budget time for install problems. Pair students with working installs with those without.

## Week 11: Code injection in memory

**Objectives**

• Find injected and unlinked modules.

• Tie a memory finding to disk and network artifacts.

**Case**

Injected Code in Memory. A workstation called a flagged address and nothing showed up on disk. Work its memory image, disk image and packet capture to find what was running, what it loaded, and where it called.

Practice, about 119 min · Up to 20 questions · Tools: Volatility, The Sleuth Kit, Wireshark, text editor

**Reading**

Volatility 3 docs: malfind, ldrmodules. One vendor write-up on process injection of your choosing.

**Before the lab**

Read the Volatility 3 pages for malfind and ldrmodules.

**Deliverable**

Lab 10 answers and the one memory finding tied to a disk or network artifact.

**Instructor note**

Hard case; set expectations that a partial but well-evidenced answer scores.

## Week 12: Network forensics

**Objectives**

• Read a PCAP in Wireshark.

• Extract hosts, sessions and a transferred file.

**Case**

First Look: Reading a Packet Capture. One packet capture from a workstation that looked compromised. Work out what it looked up, what it kept contacting, where it logged in, and what it sent.

Practice, about 33 min · Up to 15 questions · Tools: Wireshark, tshark

**Reading**

Wireshark User's Guide chapters 3 (capturing) and 6 (display filters); Follow TCP Stream.

**Before the lab**

Install Wireshark. Work the three built-in capture tutorials.

**Deliverable**

Lab 11 answers and the number of bytes pushed out over FTP, with the stream that shows it.

**Instructor note**

Display filters are the whole skill; demonstrate five before the lab.

## Week 13: Mobile forensics

**Objectives**

• Query an Android mmssms.db and recover deleted rows from the free space inside its table pages.

• Convert millisecond Unix timestamps.

**Case**

The Texts They Deleted. An employee deleted texts before handing in a company phone. Get back what was deleted and work out who it was with.

Practice, about 70 min · Up to 15 questions · Tools: DB Browser for SQLite, ALEAPP, Autopsy

**Reading**

SP 800-101r1 sections 3 and 4 (mobile data stores and extraction levels). SQLite file format documentation: b-tree pages and freeblocks.

**Before the lab**

Read the SQLite file format page on b-tree pages and freeblocks, and practice converting millisecond Unix timestamps.

**Deliverable**

Lab 12 answers.

**Instructor note**

Deleted-row recovery is the highlight; have students show the carved bytes in the page, not only the answer.

## Week 14: Capstone practical and report

**Objectives**

• Trace a stolen service account from one workstation to the server it reached.

• Deliver an investigation report with chain of custody.

**Case**

Credential Theft and Lateral Movement. A service account turned up on servers it never touches. Work out from one workstation's logs, memory and a capture how it was taken and where it went.

Timed exam, 120 min, with report · Up to 20 questions · Tools: Volatility, EvtxECmd, Wireshark, Registry Explorer, Event Viewer, PECmd

**Reading**

Re-read SP 800-86 section 3.4 and the NIJ chapter on reporting; the rubric in section 8.

**Before the lab**

Review the report rubric and the investigation report template that comes with the capstone lab. Prepare a custody form in advance.

**Deliverable**

Capstone submitted in the portal within 120 minutes; investigation report due one week later.

**Instructor note**

Run as a timed exam with the investigation report added; the report is written afterwards from the student's own saved answers.

# 8. Report rubric

Applied to the written report that follows the final timed assessment. The report is written from the student's own saved answers for their own variant, so two reports in the same class never describe the same facts.

| Criterion | Weight | Full credit looks like |
|---|---|---|
| Findings | 40% | Each finding states what happened, when, and which account or device did it, and is correct for the student's own variant. |
| Evidence | 25% | Every finding cites the artifact, the record and the tool used; hashes and paths are exact. |
| Method | 15% | The steps could be repeated by another examiner from the report alone; assumptions are stated. |
| Timeline | 10% | Events are in one normalized timezone, in order, with sources. |
| Writing | 10% | Executive summary first; plain language; no speculation presented as fact. |

# 9. Running the course in Red Herrings

1. Open Courses, pick this course, enter the class size or import the roster CSV, and press Generate the whole course. One job per week is created with the mode and time limit in the schedule.

2. Each week, open that week's job and hand out the student ZIPs or the private portal links. Portal links can be exported as a CSV for the LMS.

3. Timed weeks run in exam mode with integrity logging. Release scores from the job page after the sitting.

4. Grade from the job page or the Grading screen; export to Canvas, Blackboard, Moodle, D2L or QTI, or let LTI pass grades back.

5. Term Analytics shows the whole course at the end of term: scores by week, the questions most often missed, and integrity events.

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